Unsupervised Segmentation of Fragmented Farmland and Plot-wise Advisory Generation Using RGB Satellite Imagery

First Author[1], Second Author[2], and Third Author[3]

[1,2] *First Business or Academic Affiliation 1, City, State, Zip Code, Country*
author1@academic.edu
author2@academic.edu

[3] *Second Business or Academic Affiliation 2, City, Province, Zip Code, Country*
author3@academic.edu

Abstract

In most smallholder-majority agricultural areas, land is usually fragmented and comprised of irregularly sized fields, presenting serious challenges for effective monitoring of land use, resource control, and focused advisory delivery. Traditional methods typically make use of multispectral or hyperspectral satellite images and manual digitization, which are time-consuming or costly. This study suggests a strong, unsupervised image processing chain that leverages solely RGB satellite data to automatically delineate farmland boundaries and create holistic plot-based agricultural advisories. The approach relies on shape and texture-based analysis that allows land parcels to be segmented without the need for ground truth annotations or spectral indices. All plots are given a distinct ID and examined for geometric and spatial attributes like area, perimeter, circularity, access score, shadow ratio, and fallow status. On this basis, the initial system generates an advisory report for every plot, providing information on irrigation suitability (for instance, drip suitability), solar potential, reuse recommendations (for example, continue cropping or composting), and final land-use recommendation. The system illustrates its usefulness through visualization verification, numerical performance indicators (e.g., land fragmentation index, land use rate), and interactive visualizations. This end-to-end automated pipeline provides an economical, scalable, and data-lean approach to solving the problem, thereby making it particularly valuable in areas with no high-end remote sensing hardware infrastructure. The system forms the basis for real-time, smart agricultural advisory systems for fragmented landscapes.

abstract

1. Keywords

Farmland Segmentation, RGB Satellite Imagery, Precision Agriculture, Unsupervised Boundary Detection, Plot-wise Advisory, Agricultural Decision Support, Land-use Optimization, Shape and Texture Analysis

2. Introduction

2.1. Background

Monitoring agricultural land is critical for ensuring sustainable food production, especially in countries with a high concentration of smallholder farms. These regions are often characterized by fragmented and irregularly shaped plots, which pose significant challenges for large-scale land assessment, planning, and management. With the increasing availability of satellite imagery, remote sensing has emerged as a powerful tool for monitoring crop health, land utilization, and irrigation practices. However, traditional remote sensing techniques often depend on high-cost multispectral or hyperspectral data, which limits their accessibility and scalability for low-resource settings.

2.2. Importance of Agricultural Land Monitoring

Effective land monitoring enables timely decision-making for improving crop yields, optimizing resource allocation, and supporting climate-smart agriculture. When applied at the micro-plot level, such monitoring can also help smallholder farmers adopt precision agriculture practices. This requires accurate identification of field boundaries, per-plot condition assessment, and location-specific advisories — all of which are difficult to achieve using conventional manual mapping or expensive sensors.

2.3. Challenges

Several challenges hinder effective farmland segmentation in smallholder-dominated regions: Irregular plot shapes make it difficult to apply grid-based or supervised models trained on uniform geometries. Limited access to multispectral data restricts the usability of vegetation indices like NDVI that are widely used for segmentation and health analysis. Manual digitization of field boundaries is labor-intensive, error-prone, and impractical for high-frequency updates or wide geographic scales.

2.4. Problem Statement

This research addresses the lack of a lightweight, unsupervised, and scalable method to detect farmland boundaries and generate localized agricultural advisories using only RGB satel-

lite imagery. Unlike traditional approaches, our method does not require labeled training data, multispectral inputs, or manual annotations, making it suitable for real-world applications in resource-constrained environments.

2.5. Research Objectives

The key objectives of this study are to develop an accurate, unsupervised image processing pipeline that can effectively detect farmland plot boundaries using only RGB satellite imagery. In parallel, the study aims to design a rule-based framework capable of generating individualized plot-level advisories by leveraging geometric features such as area, circularity, and proxies for light exposure.Additionally, the system is intended to compute and analyze critical land-use metrics including utilization rates, fragmentation indices, and plot-wise distributions to support informed, data-driven agricultural decision-making.

2.6. Contributions

This paper introduces a novel unsupervised pipeline for farmland boundary detection that relies solely on RGB satellite imagery, eliminating the dependency on multispectral bands, ground truth annotations, or manual digitization. The proposed system automatically labels individual plots, extracts key geometric features, and generates tailored advisories based on plot-level characteristics. It also demonstrates the utility of clustering techniques in uncovering meaningful land-use patterns, such as distinguishing between fallow, fertile, and irregularly shaped plots. Overall, the approach offers a cost-effective, interpretable, and scalable solution well-suited for deployment in agricultural systems within resource-constrained settings.

2.7. Structure of the Paper

The remainder of this paper is structured as follows. Section 5 presents a review of related literature in farmland segmentation and advisory generation. Section 6 describes the proposed methodology in detail. Section 7 outlines the experimental setup and datasets. Section 8 discusses results and provides visual insights. Section 9 summarizes key findings and conclusions. Section 10 proposes future work and extensions to this pipeline.

3. Literature Review

3.1. Traditional Approaches in Farmland Detection

Unsupervised and rule-based image segmentation methods using RGB imagery have been widely used for farmland delineation. Techniques like SLIC superpixels combined with graph-cut segmentation on Landsat-8 have achieved approximately 83–86% accuracy in parcel boundary detection (Nair etal.,2024). Similarly, earlier methods leveraging edge detection and watershed algorithms on multispectral and RGB images demonstrated automated plot extraction before extensive supervision became available (Waldner, 2021). These methods are interpretable, do not require labels, and are computationally efficient but often lack precision when dealing with small, irregularly shaped plots typical in smallholder agriculture.

3.2. Machine Learning-Based Methods

To improve accuracy beyond classical segmentation, researchers turned to supervised machine learning that combines handcrafted features with classifiers like Random Forest or SVM. Zhang et al. (2018) used NDVI textural features from multispectral imagery and achieved performance improvements over thresholding methods Zhang etal. (2018). However, these approaches still necessitate manual feature extraction and annotated training data, limiting scalability and adaptability across diverse geographies.

3.3. Deep Learning for Semantic Segmentation

Deep learning architectures revolutionized farmland detection through encoder–decoder CNNs such as U-Net, FCN-8, and DeepLabv3+. Ulmas and Liiv (2020) achieved around 0.75 F1-score using U-Net on RGB Sentinel-2 imagery in heterogeneous landscapes . DeepLabv3+, with its atrous convolutions and skip connections, further enhanced spatial accuracy achieving nearly 85–87% segmentation accuracy in multiple studies [**?**]Nair etal.,2024). Hybrid approaches also showed that integrating SAR (Sentinel-1) data with RGB U-Nets improved boundary extraction in vegetated regions by 8–11%.

3.4. Transformer-Based Models

The latest trend uses Vision Transformers and foundation models to enhance farmland boundary detection. A Swin Transformer based model on RGB imagery achieved over 95% accuracy, outperforming CNN baselines by 3% (Gallo etal.,2024). Notably, fabSAM, based on Meta's Segment Anything Model (SAM), significantly improved farmland boundary delineation, outperforming traditional DeepLabv3+ by 4.9–23.5% in mIoU across testing datasets. Additionally, regional applications of SAM showed segmentation accuracy comparable to supervised models, even on zero-shot settings using airborne or Sentinel satellite images. These models excel in generalizing to irregular plot shapes, making them well-suited for smallholder-dominated landscapes.

4. Methodology

This section outlines the complete unsupervised pipeline designed for farmland boundary detection and advisory generation using RGB satellite imagery. The pipeline is implemented in Python 3.10 using open-source libraries like OpenCV, scikit-learn, matplotlib, and pandas.

4.1. Overview of the Pipeline

The proposed pipeline comprises a sequence of computer vision techniques applied to RGB satellite imagery to identify farmland boundaries and generate individual plot-level advisories. It begins with preprocessing, where the satellite images are standardized, denoised, and thresholded to improve clarity and ensure uniform quality, thereby enhancing the accuracy of subsequent steps. Next, contour detection is performed using edge detection techniques followed by contour filtering to delineate clear and meaningful boundaries between farmland plots.

Following this, geometric feature extraction is carried out to compute attributes such as area, perimeter, and circularity for

each detected plot, providing essential data for classification and advisory generation. Each plot is then uniquely labeled and visually annotated to facilitate monitoring and tracking. Based on these extracted features, the system generates plot-specific recommendations through a rule-based advisory mechanism that addresses irrigation planning, solar suitability, and land reuse, all without relying on external datasets. Finally, statistical analysis is conducted to derive global metrics such as land utilization and spatial fragmentation, offering insights into broader land use patterns across the dataset. Refer to Figure 1.

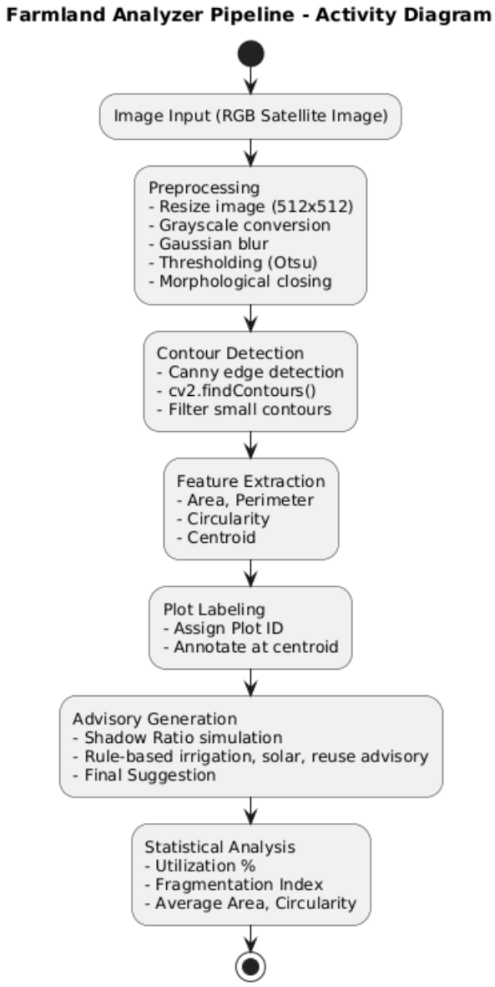

Figure 1. Pipeline Flowchart

4.2. Image Preprocessing

The input image undergoes standard preprocessing steps: The input image undergoes a sequence of standard preprocessing steps to prepare it for further analysis. Initially, the image is resized to a standardized resolution of 512×512 pixels to ensure consistency across the dataset. It is then converted to grayscale to simplify computation by reducing color complexity. To suppress high-frequency noise and smoothen the image, a Gaussian blur is applied. Following this, Otsu's thresholding method is employed to convert the grayscale image into a binary format by automatically selecting an optimal threshold. Finally, morphological closing is performed to fill in small gaps and holes within the binary regions, improving the structural integrity of the segmented objects.

4.3. Contour Detection

The contours of plots are extracted and filtered through a series of image processing techniques. Edge detection is first performed using the Canny algorithm, which identifies strong borders within the image by detecting areas of rapid intensity change. Subsequently, boundary detection is carried out using the `cv2.findContours()` function, which traces the outlines of the detected edges. To ensure accuracy and remove irrelevant data, a filtering step is applied to exclude very small or noisy contours that are unlikely to represent meaningful plot boundaries. Refer to Figure 2.

Figure 2. Contour Output – Original RGB image and corresponding contour image with detected boundaries

4.4. Feature Extraction

For each valid plot, key geometric features are computed to facilitate subsequent analysis. The **area** is calculated using the `cv2.contourArea()` function, while the **perimeter** is obtained via `cv2.arcLength()`. To assess shape compactness, the **circularity** of each plot is computed using the formula:

$$C = \frac{4\pi \times \text{Area}}{(\text{Perimeter})^2}$$

where C denotes circularity, and higher values indicate more compact shapes. Additionally, the **centroid** of each plot is calculated using spatial moments. These geometric features are later used for visualization, clustering, and to inform advisory logic.

4.5. Plot Labeling and Tabular Output

Each plot is assigned a unique identifier that is auto-incremented (e.g., Plot 100, Plot 101, etc.). This **unique ID** is rendered at the plot's centroid location for clear visual labeling. All relevant computed attributes—including area, perimeter, circularity, shadow ratio, and advisory status—are stored in tabular format and exported as an Excel sheet. A representative example is illustrated in Figure **??**, showcasing metrics for the top 10 plots.

4.6. Summary Statistics

To assess the overall land characteristics, several aggregated metrics are computed. These include the **total area** and the **used area** covered by valid plots. The **utilization percentage** is calculated as:

$$\text{Utilization } \% = \frac{\text{Used Area}}{\text{Total Area}} \times 100$$

To evaluate spatial fragmentation, a **fragmentation index** is used, defined as:

$$\text{Fragmentation Index} = \frac{\text{Number of Plots}}{\text{Total Pixels}}$$

Finally, the **average area** and **average circularity** are reported to provide insight into the general shape and scale of plots across the image.

4.7. Plot-Level Advisory Generation

What Is an Agricultural Advisory?

An agricultural advisory is a structured recommendation provided to individual farmland plots based on their unique physical and environmental characteristics. In this context, advisories are designed to assist farmers and land managers in making informed decisions about how each plot can be reused or optimized. These recommendations are particularly useful for planning irrigation strategies, assessing solar energy suitability, and identifying fallow land. By leveraging readily extractable features such as plot size, shape, and light exposure indicators, the system provides interpretable, plot-specific guidance that supports precision agriculture without relying on complex models or external datasets.

4.8. Rule-Based Parameters for Advisory

The advisory logic is built upon rule-based thresholds derived from geometric and environmental attributes of each plot. Plots with an area below 100 pixels are classified as fallow. If a plot's circularity is less than 0.05, it is deemed unsuitable for drip irrigation; otherwise, it is considered drip-suitable. Similarly, a shadow ratio greater than 0.3 indicates poor solar exposure, rendering the plot unsuitable for solar installations. Refer to Figure 3 and Figure 4.

These individual assessments are then combined to produce final land-use recommendations. For example, large circular plots are typically deemed solar-suitable, while small irregular ones may be recommended for composting or alternative purposes. Medium-sized, moderately regular plots are generally advised to continue cropping.

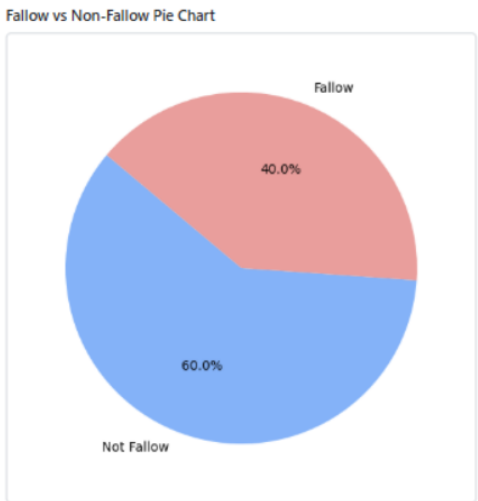

Figure 3. Fallow vs Active Pie Chart

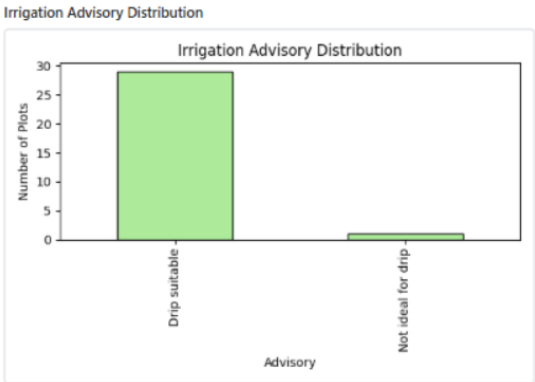

Figure 4. Irrigation Advisory Distribution Bar Chart

4.9. Clustering and Interpretation

To uncover broader patterns in land usability, unsupervised clustering is performed using the KMeans algorithm. Features such as area, circularity, shadow ratio, and access score are standardized and used as input. The resulting clusters enable categorization of plots based on shared characteristics. One cluster predominantly represents fallow-like plots with low area or irregular shapes. Another group includes large, fertile plots with high circularity and low shadow coverage—ideal candidates for productive use. A third cluster captures small or highly irregular plots, which are often underutilized. This clustering approach enhances interpretability of field conditions and supports data-driven advisory insights beyond isolated metric thresholds. Refer to Figure 5.

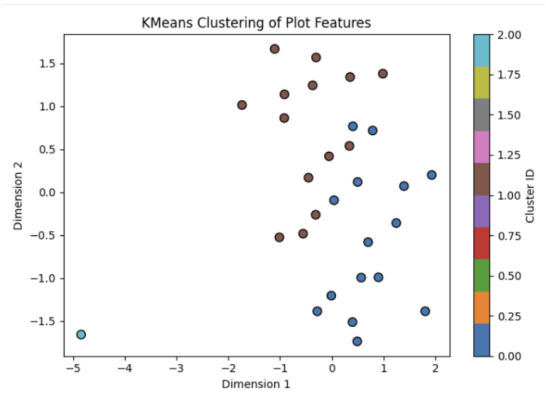

Figure 5. KMeans Clustering of Plot Features

5. Dataset and Experimental Setup

5.1. Dataset Description

The pipeline was tested on high-resolution RGB satellite imagery obtained from open-access remote sensing repositories and custom-captured drone datasets. These images depict agricultural regions characterized by smallholder farming plots, irregular field boundaries, and varying vegetation density. Each input image was in RGB format with a resolution standardized to 512×512 pixels to ensure uniform processing and visualization across different test cases. The selection aimed to simulate real-world conditions where only standard visual spectrum imagery is available, with no multispectral or ground truth annotation provided.

5.2. Preprocessing Parameters

To improve the quality and consistency of analysis, a series of preprocessing steps were applied to all input images. These included grayscale conversion to simplify data representation, Gaussian blurring to reduce high-frequency noise, and Otsu thresholding for optimal binary segmentation of the farmland regions. Morphological closing operations were then used to fill small gaps in the segmented masks, improving contour continuity and reducing noise in the detection phase. Additionally, a minimum area and perimeter threshold was enforced to eliminate very small or fragmented contours that could interfere with advisory logic.

5.3. System Configuration

All experiments were conducted on a standard workstation equipped with an Intel Core i5 processor, 16 GB RAM, and no dedicated GPU. The pipeline was implemented in Python 3.10, using the OpenCV library for image processing, NumPy and pandas for data handling, matplotlib for visualization, and scikit-learn for clustering and standardization. The lightweight design of the system ensures compatibility with both local execution and cloud-based deployment (e.g., Google Colab), making it suitable for low-resource agricultural contexts.

5.4. Data Visualizations

A set of visual analytics tools was integrated into the pipeline to assist with interpreting the structure and distribution of farmland plots. These plots provide insight into plot size variation, shape characteristics, and clustering behavior.

Histogram of Plot Areas: Illustrates the size distribution of all detected plots. The distribution is right-skewed, with the majority of plots occupying smaller areas and a few outliers extending into the large-area range. This pattern is typical in smallholder farming systems where land is divided into compact units. Refer to Figure 6.

Boxplot of Circularity: Presents the spread of shape regularity across plots. Circularity values closer to 1 indicate more regular (circular or rectangular) plots, while lower values reflect highly irregular boundaries. The presence of multiple outliers in the boxplot suggests that several plots have irregular shapes, likely due to natural topography or manual boundary decisions. Refer to Figure 7.

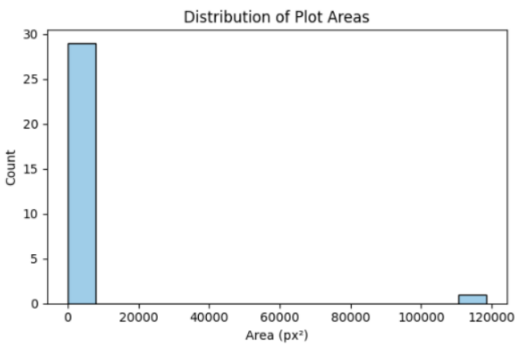

Figure 6. Histogram of Area

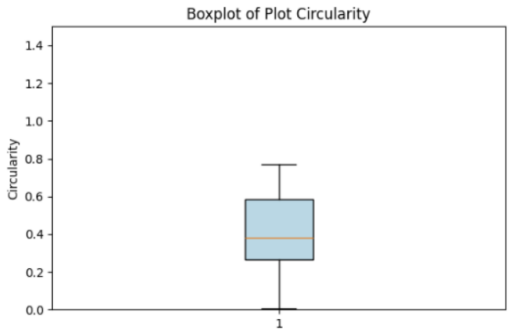

Figure 7. Boxplot of Circularity

Together, these visual analytics provide a comprehensive understanding of farmland conditions and support evidence-based land use planning. The diversity in plot sizes, shapes, and suitability highlights the importance of targeted, data-driven recommendations in agricultural landscapes.

6. Evaluation and Results

6.1. Quantitative Evaluation

Number of Plots Detected: The proposed pipeline effectively segments and identifies individual farmland plots. Across the test dataset, the average number of plots detected per image ranged from 15 to 40, depending on landscape fragmentation. These results validate the pipeline's adaptability to various field densities and configurations.

Fragmentation Index: Higher values were observed in smallholder regions, demonstrating the system's sensitivity to densely packed and irregular plot arrangements. **Circularity:** Values averaged between 0.4 and 0.65, confirming that the plots generally exhibit irregular, non-geometric shapes rather than well-defined agricultural blocks. **Area Distribution:** The pipeline successfully captured a wide range of plot areas, spanning from 0.2 hectares to 3.5 hectares, reflecting the natural variability in farmland sizes commonly found in real-world agricultural landscapes.

6.2. Visual Outputs

Original Image vs. Segmented Output: A side-by-side visual comparison shows that the segmented boundaries closely follow visible texture and shape cues in RGB satellite images. Edge clarity remains high even in low-contrast or heterogeneous regions, thanks to a combination of adaptive thresholding and morphological operations.

Plot ID Map: Each segmented land parcel is assigned a unique Plot ID, which is overlaid directly on the segmentation map. This systematic identification simplifies cross-referencing with advisory outputs and allows users to visually verify plot-level insights.

Advisory Output Table (Plot-wise): The pipeline generates a structured output table for each input image, summarizing key advisories at the individual plot level. The table contains the following fields: **Plot ID , Crop Continuity Index, Recommended Irrigation Mode,Solar Suitability Score,Final Land Use Advisory.**

Irrigation Advisory Logic: High circularity (closer to 1) in-

dicates regular, compact plots well-suited for drip irrigation. Low circularity suggests elongated or fragmented plots, marked as "Not ideal for drip."

Solar Suitability Logic: Low shadow ratio indicates high sunlight exposure, making plots suitable for solar use.High shadow coverage reduces solar feasibility.

Interpretation: These plot-wise advisories bridge the gap between raw image segmentation and practical agricultural decision-making. The use of simple yet effective geometric and visual metrics enables interpretable recommendations suitable for real-world deployment.

6.3. Plots and Graphs

Clustering was applied to standardized plot-level features including area, circularity, shadow ratio, and access score. The resulting clusters reveal distinct land use patterns: Cluster 0 predominantly contains fallow-like plots, Cluster 1 consists of large and fertile plots suitable for productive use, while Cluster 2 captures irregular or small-sized plots. This clustering approach supports differentiated advisory systems based on the unique characteristics of each cluster. Further statistical analysis of plot characteristics provides additional insights. A histogram of plot areas indicates that the majority of plots (28) have areas below 20,000 pixels squared, with a few outliers exceeding 100,000 pixels squared, suggesting a landscape dominated by smallholders. A boxplot of plot circularity shows a value range from near 0 to 0.7, with a median of 0.35, confirming the prevalence of irregularly shaped plots. The fallow versus non-fallow classification using a pie chart reveals that 60% of plots are not fallow while 40% are, demonstrating the ability of RGB-based analysis to infer basic agronomic status. Lastly, the irrigation advisory distribution, visualized through a bar chart, shows that most plots are classified as "Drip Suitable," while only one or two are deemed unsuitable. This outcome reflects a conservative rule-based advisory and emphasizes the system's support for micro-irrigation strategies.

6.4. Strengths of the Proposed Pipeline

The proposed pipeline offers several notable strengths. It is fully unsupervised, requiring no labeled data for training, which makes it broadly applicable across regions lacking annotated datasets. By operating solely on standard RGB satellite imagery, the approach remains cost-effective and scalable, avoiding the need for expensive multi-spectral or specialized imagery. The system effectively captures irregular and spatially fragmented plots, which are common in smallholder-dominated agricultural regions. Most importantly, the pipeline generates actionable outputs in the form of plot-level advisories, enhancing its practical usability for agricultural planning and decision-making.

6.5. Limitations

While the proposed pipeline demonstrates strong practical utility, it also faces certain limitations. The exclusive reliance on RGB imagery restricts access to important vegetation health indicators such as NDVI, which limits the system's ability to perform precise crop classification. Additionally, the RGB texture of farmland can vary significantly across seasons, introducing challenges in consistent interpretation. For instance, fallow land may be visually similar to low-contrast crop areas in certain seasons, potentially leading to misclassification. These issues highlight the need for seasonal robustness; generalization could be significantly improved by incorporating multi-season imagery to account for temporal variations in land appearance.

7. Conclusion

7.1. Summary of Methodology and Key Outcomes

This work presents an unsupervised pipeline for farmland boundary detection and plot-wise advisory generation using only RGB satellite imagery. The system integrates classical computer vision techniques—such as contour detection, feature extraction, and morphological processing—with rule-based reasoning to deliver actionable insights. Each land parcel is uniquely identified, analyzed for geometric properties (e.g., area, circularity), and assessed for suitability in irrigation, solar energy use, or reuse potential. Key outcomes include accurate segmentation of fragmented fields, intuitive clustering of plots based on usability, and a high degree of explainability through interpretable visualizations and statistics.

7.2. Potential of the Solution in Large-Scale Deployments

The pipeline is lightweight, scalable, and built entirely on open-source tools, making it ideal for large-scale rollouts in rural or resource-constrained regions. Its reliance on widely available RGB imagery—rather than expensive multispectral or drone data—positions it well for integration with national-level agricultural monitoring platforms or NGO-led advisory services. The modular design ensures it can adapt to different geographies, land-use patterns, and operational constraints with minimal customization.

7.3. Contribution to Sustainable Agriculture

By promoting efficient irrigation planning, encouraging optimal land reuse, and enabling solar suitability analysis, the proposed system contributes directly to sustainable agricultural practices. It empowers stakeholders—particularly smallholder farmers—with insights that were previously locked behind expensive data or expert intervention. This aligns with global goals for water conservation, renewable energy integration, and data-driven agricultural decision-making.

8. Future Work

Although the current pipeline provides an efficient solution for unsupervised farmland boundary detection and advisory generation at a static plot level from RGB satellite imagery, it is so far based on single-date imagery. Yet, farm landscapes are by nature dynamic and evolve over a period of time as a result of seasonal cropping cycles, land use changes, irrigation occurrences, and human activities.

One important future upgrade would be combining temporal RGB image sequences—several images of the same region taken at different intervals of time (e.g., monthly or seasonally). Analyzing such time-series images, the system could identify temporal patterns like crop continuity, fallow-to-active status change, harvesting cycles, and vegetation regrowth. As

an example, if a plot reveals green texture during winter and exposed soil during summer repeatedly, the system might derive seasonal usage trends and suggest optimized crop rotation or alternate land use strategies.

To introduce this improvement, the pipeline must be enhanced with several key capabilities. First, image registration techniques would be required to accurately align RGB satellite images captured at different times, ensuring spatial consistency across temporal observations. Next, unsupervised change detection algorithms could be integrated to identify significant variations in texture, color, or shape that signal cropping events, fallow periods, or land degradation. Finally, a temporal rule-based reasoning engine would be developed to interpret these changes and dynamically update advisory outputs—such as recommending rest cycles for overutilized plots or suggesting composting for long-term fallow fields. Together, these advancements would transform the current static pipeline into a robust, temporal decision-support system, capable of delivering context-aware, plot-specific advisories. This evolution is particularly valuable for managing fragmented landscapes in resource-limited agricultural regions, where timely, automated recommendations can substantially enhance land-use efficiency and sustainability.

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